POWER INPUT & PROTECTION

J1
Pin_1
Pin_2
VIN_RAW
24 V DC input
VIN_FUSED
GND
100nF 50V
GND

POWER REGULATION

Q1
VIN_PROT
VIN_FUSED
Q1_GATE
S1 S2 S3 G D1 D2 D3 D4
IRF7240PBF

U1
VIN_PROT
VCC5
FB5
BUCK5_BST
FB BOOT VIN EN VCC PG SW PGND PAD
PWR_GOOD_N
BUCK5_SW
LMR36510UADDAR
GND

U2
+3V3
+5V
VO VI GND
AMS1117-3.3
GND

D1 VIN_FUSED 100nF 100V +3V3_A GND FLG3 +3V3_A
D2 VIN_FUSED SMBJ24CA VIN_PROT GND FLG4 VIN_PROT
D3 VIN_FUSED Q1_GATE MMBZ5242B
VIN_RAW VIN_FUSED
Q1_GATE GND 10R
C5 +3V3_A 10nF 5V_PWR GND
C6 VIN_PROT 100nF 50V PWR5_LEDA GND
C7 VIN_PROT 100nF 100V PWR3_LEDA GND
C8 VIN_PROT 22nF 50V +5V FB5_TOPA GND
C9 BUCK5_BST 10nF BUCK5_SW GND
C10 VCC5 1u +3V3 GND
L1 BUCK5_SW 3.3uH 1.5A BUCK5_SW PWR_GOOD_N
R1 FB5_TOPA 47.5k FB5
R2 FB5 20k GND
R3 +3V3 PWR_GOOD_N 10k
FLG1 GND GND
C20 +5V 100nF +5V GND FLG2 +5V
C21 +5V 22uF 10V +3V3 GND
C22 +3V3 100nF +3V3_A GND
C23 +5V 22uF 10V +5V GND
C24 +3V3 100nF PWR5_LEDA GND
C25 +5V 22uF 10V +3V3 GND
C26 +5V 100nF PWR3_LEDA GND
C27 +5V 100nF FB5_TOPA GND
C28 +5V 100nF GND

MISC

FLG5 +5V_ENC
FLG6 +3V3_SD
FLG7 +5V_ISO_IN
FLG8 +12V_FIELD

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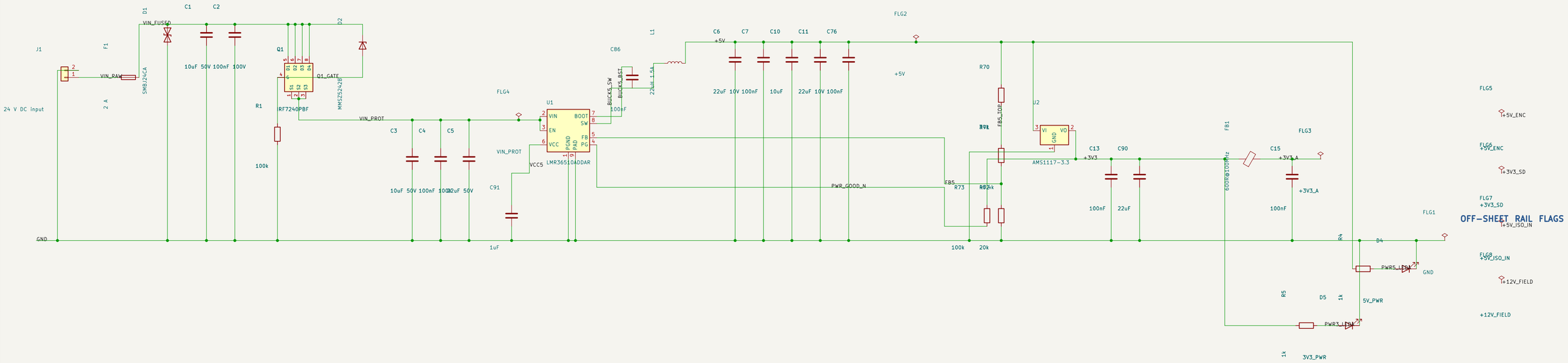
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Size: A3	Date: 2026-08-10	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/1

Power flow: 24 V input -> fused/protected VIN -> 5 V buck -> 3V3 LDO

INPUT • FUSE • REVERSE-POLARITY PROTECTION

5 V BUCK POWER STAGE

3V3 LDO • FILTER • STATUS



Method B: existing symbols, explicit orthogonal wiring

LeanCam / uCNC experiment

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Title: LeanCam NCC RP2350B – Power and Supervision (Python re-layo

Size: A2	Date: 2026-10-29	**Rev:**
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